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Original Article

Renoprotective effect of *Hemidesmus indicus*, a herbal drug used in gentamicin-induced renal toxicity

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SUMMARY:

Background and Aims: Owing to the global trend towards improved 'quality of life', there is considerable evidence of an increase in demand for medicinal plants. The WHO guidelines define basic criteria for the standardization of herbal medicines. The present work is an effort in this direction to prove the safety and efficacy of *Hemidesmus indicus* Linn. in the management of nephrotoxicity induced by aminoglycosides such as gentamicin.

Methods and Results: Simple, quality control methods using high performance thin layer chromatographic (HPTLC) phytochemical fingerprint, proximate analysis, and the stability of the *H. indicus* root powder were developed. From the toxicity study using albino Swiss mice, it was observed that the drug (*H. indicus*) was relatively safe up to 7 g/kg bodyweight dose. Efficacy was evaluated against gentamicin-induced nephrotoxicity in albino Wister rats. The study examined animals from the following groups: no treatment, gentamicin treated, gentamicin treated recovery, and gentamicin and plant treated. Animals from all groups were killed on day 13 of the study; those from gentamicin treated group were killed on the seventh day. Assessment of the drug efficacy drug was conducted by using haematological and histological examination.

Conclusion: The treatment with *H. indicus* helped in the management of renal impairment, which was induced by gentamicin in rats. This is evident from the results obtained for various kidney function tests for gentamicin, along with the results from the plant treated group, and is in comparison with the results found for the gentamicin recovery group. A histological examination of kidneys also supports the findings from haematological evaluations. The plant shows promise as an adjunct therapy along side aminoglycosides as it reduces nephrotoxicity caused by aminoglycosides.

KEY WORDS: **aminoglycosides, gentamicin, *Hemidesmus indicus*, indigenous medicine, nephrotoxicity, rats.**

INTRODUCTION

India has one of the world's most sophisticated indigenous medical cultures, with an unbroken tradition that has spanded over more than four thousand years. Even today, millions of people, in villages, towns and cities across the country, depend upon it for their health-care needs. India also has rich biodiversity and a treasure of medicinal plants because of the diverse agroclimatic conditions. This resource is becoming scarce day-by-day because of over exploitation and unscientific harvesting of medicinal plants.

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Increasing health concerns today necessitate reliance on alternate remedies for sustaining good health. Owing to the global trend toward improved 'quality of life', there is considerable evidence of a phenomenal demand for medicinal plant-based supplements from natural sources that have had no contamination from synthetic fertilizers or chemicals.

For regulating quality of herbal medicines, a different approach is needed than that used for synthetic drugs. The single and most important factor that stands in the way of a wider acceptance of traditional herbal medicines is the non-availability or inadequacy of standards for checking their quality by chemical or bioassay methods. Scientifically generated data on herbal medicines will not only help cure the people in the world but will also help to sustain the demand in the global market.

Thus, standardized drugs of well-defined consistent quality are needed for reliable clinical trials and therapeutic use.

The major reason put forth for the difficulty in developing quality control standards is that these products use whole plants or parts of plants or their total extracts, and in some cases, even a mixture of a number of plants. Standardization of herbal drugs and of a preparation there of, is not just an analytical operation, because it also embodies total information and controls, which are necessary to guarantee consistency of composition. Therefore, the World Health Organization (WHO) has stressed the importance of the standardization of plant drugs. The objective of the present research was to develop appropriate techniques to study the herb, *Hemidesmus indicus* Linn (family: Asclepiadaceae), so that they are in line with the guidelines suggested in the classified Ayurvedic Pharmacopoeia, Indian Herbal Pharmacopoeia, Indian Council of Medical Research (ICMR) guidelines and WHO monographs.

Hemidesmus indicus, also called as 'Sariva', is being used in traditional medicines for the treatment of various diseases such as diarrhoea, rheumatoid arthritis and urinary disorders. According to the Ayurvedic textbook, the plant, therapeutically is an alterative, diaphoretic, diuretic. The plant is also used in the treatment of constitutional debility and in respiratory disorders. It is also used as an antisyphilitic and as a galacogogue.[1,2] The plant has been reported to be useful in the management of fever[3] and in the treatment of tumours.[4] It has also been proven to be an immunomodulatory agent,[5] and useful in viper venom neutralization.[6] A viper venom inhibitory factor from the root of the plant has been isolated, purified and partially characterized.[7] The plant also has been reported to be used in ophthalmic formulations in combination with other herbs.[8] It is also used by tribes of India in hair and scalp preparations.[9]

The plant has been shown to have antiviral activity and was found to be effective in inhibiting the replication as well as cytopathic activity of vaccinia and Ranikhet viruses when various plant extracts obtained from *H. indicus* cultures were tested in cell cultures infected with vaccinia and Ranikhet disease viruses.[10] The extract derived from the cell culture of *H. indicus* roots was found to be effective as it prevented hypercholesterolaemia in experimental rats and there was also a significant reduction of lipid levels.[11] The plant oil has been reported to exhibit antimicrobacterial activity; the extracts studied were petroleum ether (60–80), chloroform, and alcohol.[12] The above extracts have been reported to be effective against *Salmonella typhosa*, *Vibio cholera*, *Staphylococcusaureus*, *Escherichia coli*. The plant has shown protective action against rifampicin and isoniazid-induced hepatotoxicity in rats.[13] Coumarinolignoids were isolated from the plant as naturally occurring compounds with cytotoxic and hepatotoxic properties.[14] Other phytochemicals such as terpenoids, β-sitosterol, pregnane glycosides, pregnane ester glycoside, and indicusin were isolated from the root powder.[15–20] 2-Hydro2-Methoxybenzoate was isolated during root culture of the plant.[21] An evaluation of the inhibitory extracts against 3-Keratinophilic fungi has been conducted using this plant.[22] The present study reports the efficacy of the root powder of *H. indicus* in managing gentamicin-induced nephrotoxicity in rats. The present study also reports the routes of standardization of the plant using various analytical techniques. Also, quality control methods using proximate and phytochemical analysis, and high performance thin layer chromatography (HPTLC) fingerprinting have been developed using *H. indicus*. Using *H. indicus*, toxicity studies on mice and efficacy study on rats were carried out by using haematological and histopathological parameters.

METHODS

Plant authentication, collection, drying and storage

Herbariums of *H. indicus* Linn were authenticated by the National Institute of Science Communication (NISCOM), New Delhi, India. They were assigned voucher number PID – 1821. The roots of *H. indicus* were collected in bulk from three regions of India; Mumbai and Melghat (Maharashtra state) and Trichur (Kerala state) during the summer, rainy and winter seasons. The roots were then washed and oven-dried and maintained at 45 ± 2°C for 3 days. The dried roots were grinded to a powder in an electric grinder and the powder was sieved through a BSS mesh number 85 sieve (Jayant Test Sieves; Jayant Scientific Ind., Mumbai, India). The sieved powder was stored in airtight, high-density polyethylene containers before analysis.

Quality control methods

Proximate analysis

Tests for moisture content, total ash, acid insoluble ash, crude fibre content, water-soluble extract, and alcohol soluble extract were carried out.[23] The total ash amount was subjected to an elemental analysis for calcium, magnesium, sulphate, chloride and iron.[24]

Optimization of extraction conditions

Extraction conditions were optimized to get the maximum percentage yield of extract by using different solvents such as water, methanol, acetonitrile and petroleum ether (60–80). The following conditions were optimized: the volume of solvent required, time of extraction, and cycles of extraction to give maximum yield extract.

Qualitative and quantitative estimation of phytochemicals

Aqueous and alcoholic extracts were analysed by the use of qualitative tests to confirm the presence of alkaloids, carbohydrates, glycosides, phytosterols, fixed oil, fats, proteins, amino acids, gums and mucilages.[25] Quantitative tests were carried out on plant root powder to calculate the percentage of alkaloids, flavenoids, fatty acids, tannins and phenolics. Visible colour changes in the root powder under UV and visible light were noted when treated with different reagents.

Chromatographic fingerprint using HPTLC

High performance thin layer chromatographic fingerprint was developed to identify the root powder of *H. indicus*, which can be used in a quality control method for the raw material. The chromatographic system includes CAMAG (Muttenz, Switzerland), Linomat IV applicator with CAMAG Video Store and a Video Scanner III for densitometric evaluation, an IBM computer with Pentium III processor; and CAT'S 4.0 version software supplied by CAMAG. The chromatographic conditions include a mobile phase (ethyl acetate : methanol : water in the volume ratio of 100:13.5:10); where the mode of separation was in the normal phase. The stationary phase was silica gel $60F_{254}$, precoated on an aluminium sheet. The spotting volume was 10 μL, the development chamber was a CAMAG Twin trough and the tank saturation time was 1.50 h. Detection was carried out under a fluorescence mode of 366 nm. The methanolic extract of the *H. indicus* root powder was used for phytochemical fingerprinting.

Sample preparation for HPTLC

One gram of root powder was weighed and transferred to a stoppered conical flask. To it, 10 cm of methanol was added and the contents of the flask were mixed properly.[3] The solution was allowed to stand undisturbed for 4 h. The solution was then filtered through Whatman filter paper number 1. The clear filtrate obtained was the cold methanolic extract. Methanolic extract from 1 g of root powder was also obtained through the soxhlet extraction process over a period of 8 h. Both the extracts were analysed by using the optimized chromatographic conditions as described above. Cold methanolic extracts of the root powder collected from three different regions and during different seasons were prepared and similarly analysed.

Stability test for the methanolic extract

The cold extract of *H. indicus* was subjected to a stability test. The extract was analysed immediately after extraction, and also at 24, 72 and 96 h and subsequently after 7, 14 and 21 days post extraction. The extract was stored in a refrigerator at 4 ± 2°C during the course of the experiment.

Toxicity study

A study relating to the determination of the median lethal dose (LD_{50}) was performed by using different doses of the *H.indicus* root powder on albino mice (weighing between 20 and 25 g), as per the Organisation for Economic Cooperation and Development (OECD) guidelines. They were maintained under standard maintenance conditions and 12-h light and dark cycles. The animals were housed in polyurethane cages and were provided with drinking water and standard AMRUT (Chakan Oils Mills Ltd, Pune, India) feed. The animals were divided into three groups, and root powder in the form of aqueous slurry was administered through oral gavage in a single dose. The animals received a maximum of 5 g/kg dose. The animals were observed for a period of 14 days post dose. Daily observation of bodyweight changes, food and water consumption, behavioural changes, and mortality (if any) were recorded.

Efficacy study

The renoprotective activity of *H. indicus* root powder was studied by using albino Wister rats of both sexes, which were treated with gentamicin. The animals (weighing between 180 and 200 g) were obtainted from Haffkine Biopharmaceutical (Mumbai, India) and were maintained under standard maintenance conditions as described previously. All processes were approved by the Medical Centre Animal Ethics Committee of the college laboratory, which is certified by the Committee For The Purpose Of Control and Supervision Of Experiments On Animals, India (CPCSEA); certificate number 135.

Drug and chemical treatment

Animals were segregated into four groups of 12 animals (six males and six females). The animals in group 1 served as the control group, which received no treatment and were killed on the 13th day after commencement of the study. Group 2 animals were the gentamicin-treated control group, which received an intramuscular (i.m.) injection of gentamicin sulfate (90 mg/kg) daily for 6 days and were killed on the seventh day. The animals in group 3 were treated with gentamicin sulfate as they were in group 2, and from seventh day onwards, the animals received an oral dose of 2 mL distilled water, per rat, for the next 6 days. The animals were killed on the 13th day of treatment. This group was the gentamicin-treated natural recovery group. In the case of group 4, gentamicin sulfate was injected again daily for 6 days, and from the seventh day, the *H. indicus* root powder was orally administered daily as aqueous slurry through a gavage. The animals were killed on the 13th day of treatment. This group was the gentamicin and plant-treated group. Daily observations of bodyweight changes, food and water consumption, behavioural changes, and mortality (if any) were recorded.

Biochemistry

Urine (24 h collection using metabolic cages) and serum (after the rats were killed) specimens were collected from rats in the normal, the gentamicin and plant-treated group, and the gentamicin treated natural recovery group on the 13th day, and from the gentamicin control group on the seventh day. Serum concentrations of creatinine, blood urea nitrogen (BUN), gamma glutamul transpeptidase (γGT), potassium, sodium, chloride, aspartate transferase (AST), alanine transferase (ALT), alkaline phosphatase, bilirubin, albumin, cholesterol and triglycerides (fasting), and urine proteins concentrations were estimated by using a STAT Fax: 2000 Autoanalyser (Merck Limited India, Mumbai, India).

Haematology

Blood cell counts were performed by using an automated cell counter on heparinized blood collected after the rats were killed.

Histology

Samples of kidney tissue were excised and rinsed in 0.9% saline and blotted dry of saline and excess blood, and were then weighed. They were fixed in aqueous Bouin's fixative (picric acid : formalin : acetic acid in the volume ratio of 75:25:5) for 24 h. The tissues, after fixation, were washed in water to remove excess fixative. Washed tissues were then dehydrated through a graded series of ethyl alcohol, cleared with xylene and embedded in paraffin. Sections were cut at 7 μm and mounted on clean glass slides. The sections were stained routinely with haematoxylin and eosin. The stained slides were observed and photographed. Prints of the 35-mm negative were made so as to obtain a sixfold magnification of the original microscope enlargement.

Statistical analysis

Data are presented as mean ± SD. Comparisons were made between the gentamicin natural recovery group and the gentamicin and plant-treated group at different time points by using the two-sided Student's *t*-test. *P* values of less than 0.01 were considered to be statistically significant.

RESULTS

Quality control methods

Proximate analysis

Analysis of *H. indicus* quantitative standards was carried out in triplicate and the mean of the three was taken. Each season and different regions were taken into consideration. The moisture content of the root powder collected in the monsoon season was found to be 15.2%, which was more in comparison with the moisture content of the root powder collected in the summer and winter seasons (9.1 and 7.1%, respectively). Similarly, acid insoluble ash, which was estimated for the summer, monsoon and winter seasons was found to be not more than 15.1, 14 and 16.2%, respectively. The crude fibre content of the root powder collected in the summer, monsoon and winter seasons was found to be 17.1, 16.4 and 19.7%, respectively. Also, the total ash, which was estimated for the summer, monsoon and winter seasons was found to be not more than 5, 4.5, and 3.6%, respectively. Elementary analysis was carried out on the root powder ash, and the calcium content was found to be not more than 25.2%, magnesium not more than 9.7%, and iron not more than 1.4%. Sulfur was found only in trace amounts. Alcohol-soluble extract was found to be not less than 1.9% for both the summer and monsoon seasons and 2.3% for the winter season. Similarly, water-soluble extract was estimated for summer, monsoon and winter seasons and was found to be not less than 18.1, 17.9 and 19.2%, respectively.

Optimization of extraction conditions

The values of extraction for different solvents is tabulated, where water, methanol, petroleum ether, acetonitrile showed 17.5, 11.95, 4.89, 3.89%, respectively (Table 1).

Qualitative and quantitative estimation of phytochemicals

Preliminary phytochemical screening showed the presence of alkaloids, carbohydrates and glycosides, phytosterols, fixed oils, fats, proteins, amino acids, gums and mucilages (Table 2). The observations were the same for the root powder of all the three seasons and regions. The above tests were further evaluated for their percentage

Table 1 Per cent extraction yield in different solvents

Solvents	Volume(mL)	Time(h)	No. cycles of extraction	% Extraction (mean ± SD)
Water	150	1.00	5	17.51 ± 0.47
MeoH	100	1.50	4	11.95 ± 0.37
ACN	100	2.50	5	3.89 ± 0.29
Petroleum ether (60–80)	75	1.50	5	4.89 ± 0.69

ACN, acetonitrile; MeOH, methenol.

Table 2 Preliminary phytochemical screening of *Hemidesmus indicus*

Extracts	Compounds	Tests/Reagents
Alcoholic	Alkaloids	Drangendroff' and Mayer's
Aqueous	Carbohydrates and glycosides	Fehling's Soln
Aqueous	Carbohydrates and glycosides	Fehling's Soln
Alcoholic	Phytosterols	Libermann's test
Petroleum ether	Fixed oils and fats	Spot test
Alcoholic	Saponins	Foam test
Aqueous	Saponins	Foam test
Alcoholic	Proteins and aminoacids	Biuret test
Aqueous	Proteins and aminoacids	Biuret test
Aqueous	Gums and mucilages	Alcoholic precipitation

Table 3 Quantitative analysis of *Hemidesmus indicus* root powder during different seasons

	Summer season	Monsoon season	Winter season
Alkaloids	6.70 ± 0.08	7.25 ± 0.02	7.51 ± 0.03
Acidic flavenoids	5.32 ± 0.02	5.34 ± 0.02	5.5 ± 0.01
Basic flavenoids	0.81 ± 0.01	0.83 ± 0.01	0.91 ± 0.01
Fatty acids	3.81 ± 0.01	3.81 ± 0.01	4.2 ± 0.01
Penolics	1.16 ± 0.01	1.20 ± 0.01	1.22 ± 0.01
Tannins	0.3 ± 0.01	0.27 ± 0.02	0.3 ± 0.01

Results are presented as mean percentage ±SD.

Table 4 Reactions of reagents with *Hemidesmus indicus* root powder

Chemicals/Reagents	Colour change
Concentrated HNO_3	Brown
Concentrated H_2SO_4	Black
$CaCl_2$	Light green
Oxalic acid	Brown
NaOH	Brown
Iodine	Dark brown
Dragendroff reagent	Orange
$FeCl_3$	Greenish yellow
$AgNO_3$	Brown
Picric acid	Yellow
Biurate	Green
Florescence	Yellow
Powder with dilute solution of ammonia	Yellowish brown

$AgNO_3$, silver nitrate; $CaCl_2$, calcium chloride; $FeCl_3$, ferric chloride; HNO_3, nitric acid; H_2SO_4, sulfuric acid; NaOH, sodium hydroxide.

evaluations. Quantitative analysis of alkaloids, flavonoids (acidic and basic), fatty acids, tannins, and phenolics of different seasons from differerent geographical regions was carried out (Table 3). A change in the colour of the root powder after treatment, when mixed with different reagents and observed under UV light, was recorded (Table 4).[25]

Chromatographic fingerprint using HPTLC

High performance thin layer chromatographic fingerprint of the root powder showed eight spots (Table 5). A typical chromatogram is depicted in (Fig. 1). An additional spot at R_f 0.26 was observed in the plant samples collected during the summer season from Melghat and Trichur, and in the samples collected during the winter and monsoon seasons from Mumbai, Melghat and Trichur. This spot was present for the plants collected in Mumbai in the winter and rainy seasons, although the intensity of the spot observed was comparatively less. Also, the spot at R_f 0.37 was absent for the sample of plants collected from Trichur in the monsoon season; this was seen in all the other plants from other regions

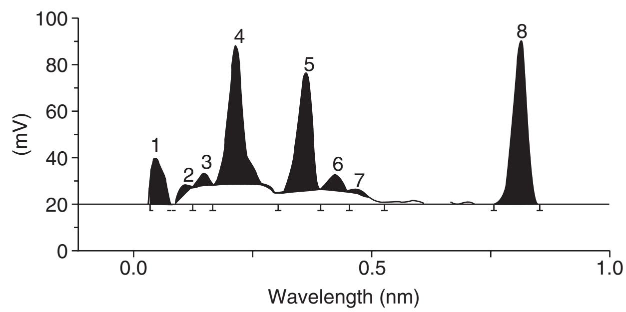

Fig. 1 A typical high performance thin layer chromatogram (HPTLC) of methanolic extract of *Hemidesmus indicus* root powder.

collected during the different seasons. The spot at R_f 0.47 was absent in the plants collected from Melghat in the rainy season and in the plants collected from Mumbai in the winter season. The plants collected from Trichur during all seasons showed the presence of this spot. The plants from the Mumbai region showed the presence of this spot in the summer and monsoon seasons, whereas plants from the Melghat region showed the presence of the spot in the summer and winter seasons.

Stability test for the methanolic extract

The stability of the methanolic extract was evaluated at the prominent peak at R_f 0.82. The results revealed that the extract was stable for 2 days and the process of degradation commenced from the third day onwards. The degradation was not significant, with the concentration of the components still present up to the fourth day. Significant degradation was observed from the first week onwards, and by the end of third week, the concentration of the three phytoconstituents was almost negligible (Table 6).

Toxicity study

On the basis of the toxicity study, it was observed that the plant extract was non-toxic and did not induce death

Table 5 Geographical and seasonal variations of *Hemidesmus indicus*

	Summer			Rainy			Winter		
R_f	Mu	Me	Tr	Mu	Me	Tr	Mu	Me	Tr
0.05	429	714	219	404	366	333	181	357	276
0.11	58	63	26	58	37	27	24	33	21
0.15	88	101	58	111	64	60	53	38	56
0.21	1629	1240	570	1114	687	581	436	584	556
0.26	ND	55	60	13	96	73	36	88	37
0.37	1358	1133	499	982	507	ND	372	471	473
0.43	165	159	51	122	66	48	27	51	59
0.47	39	60	20	41	ND	21	ND	20	17
0.82	1831	1732	656	1345	687	731	354	757	695

All values are presented as area under the curve (AUC) after a densitometric scan (366 nm). Me, melghat; Mu, Mumbai; ND, not detected; R_f, relative front; Tr, Trichur.

Table 6 Stability of the methenolic extract of *Hemidesmus indicus*

Day(s)	R_f 0.82 (AUC)
1	1826.66 ± 1.52
2	1828.33 ± 3.05
3	1786.33 ± 5.13
4	1728.33 ± 3.51
7	267.33 ± 15.53
14	96.66 ± 11.06
21	33.23 ± 6.53

Results are presented as mean ± SD. AUC, area under the curve; R_f, relative front.

at the highest possible single dose, 5 g/kg. There were no changes observed with respect to their general behaviour, food and water consumption or bodyweight changes.

Efficacy study

General appearance, behaviour and consumption

No significant changes in the general behaviour of the animals from all dose groups was observed. No animal died in the experiment. The water intake by the animals was reduced in group 2 (as compared with group 1); they received gentamicin i.m. daily for 6 days. This reduction was seen until the sixth day of the study for both sexes. For both sexes of rats in group 3, which were administered gentamicin for 6 days and were kept under observation for a further 7 days, a reduction in water intake up to the sixth day was observed; this gradually increased to normal levels by the 13th day. For the rats in group 4, which received a gentamicin injection and root powder treatment, a similar trend of water intake (to that observed for group 3) was found. Rats in group 1 did not show any changes in terms of food intake over the experimental period. No major change in food intake was observed for rats of both sexes from groups 2 and 3, but for group 4 animals, there was decrease in food intake from the seventh day onwards, but it did not decrease to the predose level. There was a decrease in the liver to bodyweight ratio for groups 2, 3 and 4 as compared with group 1 rats. The kidney to bodyweight ratio showed an increase in all the groups as compared with group 1 animals. A reduction in this ratio was found for group 4, when compared with group 3 rats.

Biochemistry

The results indicate that gentamicin increases serum creatinine and urea concentrations as compared with normal control group animals (Table 7). There was an increase in serum triglycerides, total cholesteral, γGT, bilirubin, aspartate aminotransferase (AST), alanine aminotransferase (ALT) and alkaline phosphatase in the blood levels in animals from groups 2 and 3 as compared with the normal control group 1 (Table 7). After oral administration of root slurry, the plasma levels of urea, creatinine, γGT, bilirubin, AST, ALT, and alkaline phosphatase were found to be decreased in the gentamicin and plant-treated group (Table 7). The levels of serum potassium, chloride, sodium, total protein, and albumin were not found to be different when the comparison was made between the gentamicin-treated and normal control groups.

Haematology

A decrease in the haemoglobin level of peripheral blood was seen in the animals from group 2. No significant differences in red blood cell, white blood cell, neutrophil, monocyte or lymphocyte counts were seen between normal and gentamicin-treated groups (Table 7).

Proteinuria

Controls did not show different levels of albuminuria, occult blood, pus cells and red blood cells during the experimental period. Heavy albuminuria was observed in groups 2 and 3 when compared with group 4 (2.25 ± 0.96, 2.33 ± 0.58 *vs* 1 ± 0 for gentamicin control, gentamicin natural recovery group *vs* gentamicin and plant-treated group of male rats at the termination of study, respectively; $P < 0.001$, Table 8). Rats from groups 2 and 3 showed the presence of Occult blood, pus cells and red blood cells in both male and female rats; group 4 rats did not show the above.

Table 7 Effect of gentamicin and root powder of *Hemidesmus indicus* on organ to bodyweight ratio and blood biochemical parameters in male and female rats

Test	Normal control	Gentamycin control	Gentamycin recovery	Gentamycin and root slurry
Male rats				
Liver/bodyweight ratio	0.03 ± 0.0054	0.026 ± 0.0023	0.029 ± 0.0032	0.029 ± 0.0026
Kidney/bodyweight ratio				
Rt	0.003 ± 0.0006	0.003 ± 0.0002	0.003 ± 0.0003	0.003 ± 0.0002
Lt	0.003 ± 0.0006	0.003 ± 0.0004	0.0035 ± 0.000	0.0034 ± 0.000
Haemoglobin	16.3 ± 0.47	13.3 ± 1.90	15.8 ± 1.55	17.0 ± 1.60*
RBC	6.3 ± 0.33	4.7 ± 0.65	5.6 ± 0.63	5.9 ± 0.57
WBC	6240 ± 324	6720 ± 156	6583 ± 905	5866 ± 688
Neutrophils	54.6 ± 5.27	68.5 ± 1.97	63.2 ± 8.9	51.8 ± 11.5
Lymphocytes	38.8 ± 3.9	42 ± 8.69	29.6 ± 15.19	25.6 ± 2.06
Eosinophils	3.25 ± 2.2	11.3 ± 5.77	6.25 ± 2.22	4.83 ± 1.72
Serum creatinine	0.6 ± 0.13	1.0 ± 0.16	0.73 ± 0.10	0.69 ± 0.10*
BUN	16.4 ± 2.30	21.1 ± 2.84	18.1 ± 2.40	18 ± 1.6*
TC	86.8 ± 5.06	98.8 ± 4.58	117.1 ± 11.63	110 ± 20.2
SGOT (AST)	39.8 ± 5.23	81 ± 4.65	64.5 ± 4.09	52.5 ± 3.39
SGPT (ALT)	56.9 ± 4.31	78.8 ± 8.68	67.1 ± 5.52	54.1 ± 5.22
γGT	7.83 ± 1.94	28.17 ± 11.43	20 ± 6.26	7.83 ± 1.94*
AP	73.33 ± 16.84	220 ± 44.11	159 ± 21.59	147.83 ± 24.57
Triglyceride	86.8 ± 5.06	191.5 ± 6.59	138.8 ± 4.83	116.6 ± 9.02*
Serum billirubin	0.8 ± 0.14	0.98 ± 0.15	0.77 ± 0.14	0.73 ± 0.12
Female rats				
Liver/bodyweight ratio	0.03 ± 0.0044	0.025 ± 0.0021	0.031 ± 0.0029	0.031 ± 0.0029
Kidney/bodyweight ratio				
Rt	0.003 ± 0.0004	0.002 ± 0.0003	0.0042 ± 0.000	0.003 ± 0.0001
Lt	0.003 ± 0.0004	0.002 ± 0.003	0.004 ± 0.0006	0.003 ± 0.0002
Haemoglobin	14.2 ± 0.70	12.8 ± 1.21	16.6 ± 0.82	17.1 ± 0.74*
RBC	5.12 ± 0.30	4.57 ± 0.5	5.92 ± 019	5.9 ± 0.35
WBC	1820 ± 729	4983 ± 221	4233 ± 148	2733 ± 157
Neutrophils	58 ± 2.45	69.17 ± 10.94	49 ± 11.40	51.6 ± 9.18
Lymphocytes	38.5 ± 2.9	47.2 ± 12.4	41.8 ± 12.5	24.6 ± 9.07
Eosinophils	3.75 ± 1.7	6.75 ± 3.40	6.5 ± 0.70	5 ± 1.58
Serum creatinine	0.7 ± 0.16	1.7 ± 0.26	0.97 ± 0.11	0.8 ± 0.2*
BUN	18.5 ± 1.48	21.5 ± 4.81	17.1 ± 0.70	17 ± 1.17*
TC	135.8 ± 11.4	135 ± 12.82	134.5 ± 20.8	108.1 ± 7.30
SGOT (AST)	38.91 ± 4.56	63 ± 5.96	56 ± 2.89	38.7 ± 4.27
SGPT (ALT)	44.8 ± 4.12	68 ± 8.39	57.5 ± 3.61	47.93 ± 3.79
γGT	8.0 ± 3.32	40.83 ± 4.44	22.17 ± 15.25	15 ± 4.86*
AP	70 ± 34.88	207.33 ± 11.34	191.17 ± 30.15	180 ± 38.07
Triglyceride	110.6 ± 12.13	204 ± 7.24	203.3 ± 15.32	110.6 ± 12.13*
Serum billirubin	0.7 ± 0.13	0.78 ± 0.15	0.87 ± 0.20	0.5 ± 0.10

All the values are expressed as mean ± SD, where $n = 6$. Differences, which are significant, are marked as: $^*P < 0.5$. AP, alkaline phosphate; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BUN, blood urea nitrogen; γGT, gamma glutamultransferase; LT, left; RBC, red blood cells; Rt, right; TC, total cholesterol; SGOT, serum glutamic oxaloacetic transaminase; SGPT, serum glutamic pyruvic transaminase; WBC, white blood cells.

Table 8 Effect of gentamicin and root powder of *Hemidesmus indicus* on the urine composition of male and female rats

Parameters		Normal control	Gentamycin control	Gentamycin recovery	Gentamycin root slurry
pH	Male	7.35 ± 0.56	7.67 ± 0.60	8.17 ± 0.29	8 ± 0.00
	Female	7.23 ± 0.45	6.33 ± 0.82	8.17 ± 0.29	4.37 ± 3.28
Albumin	Male	Absent	2.25 ± 0.96	2.33 ± 0.58	1 ± 0
	Female	Absent	2 ± 1	1.67 ± 1.15	1.5 ± 0.70
Occult blood	Male	Absent	3.2 ± 0.84	3.67 ± 0.58	Absent
	Female	Absent	3 ± 0.70	1 ± 0	2 ± 0
Pus cells	Male	Absent	3 ± 0	2.25 ± 0.5	Absent
	Female	Absent	2.25 ± 0.5	Absent	Absent
RBC	Male	Absent	3 ± 0	2.67 ± 0.58	Absent
	Female	Absent	2.25 ± 0.5	Trace	2 ± 0
Casts/cysts	Male	Absent	Absent	Absent	Absent
	Female	Absent	Absent	Absent	Absent

Results are presented as mean ± SD of six values. RBC, red blood cells.

Histology

Macroscopical appearance of the kidneys

Kidneys damaged by gentamicin were slightly swollen and this was similar to observations made by Rouiller.[26]

Microscopic changes

- *Gentamicin control group*: Parenchyma revealed varying degrees of congestion and moderate haemorrhages. Tubular epithelium revealed varying degenerative changes including necrosis. Many foci appeared ruptured, with remnants in the lumen. Few areas showed swollen tubular epithelium; it obliterated the lumen. Aggregation of mononuclear cells with hyperchromatic nuclei was seen around the blood vessels. Glomeruli were swollen and revealed increased cellularity. A few foci at the corticomedullary junction appeared atrophic. In this group, the infiltration (mononuclear aggregation), in general, was very mild or absent, and necrosis and cellularity were graded as being moderate, rarely as mild (Fig. 2).
- *Gentamicin natural recovery group:* The changes in the tubular epithelium and glomeruli followed a similar pattern as that observed in the above group, but they were of a moderate intensity, which was rather severe, even after 7 days of natural recovery (Fig. 3).
- *Gentamicin and plant-treated group*: The pattern of lesions was graded as being mild to moderate in the tubules and glomeruli; at certain foci, the changes were mild with less cellularity and less swelling in the epithelium (Fig. 4).
- *Normal control group*: The appearances of the kidney tissues from control rats were in accordance with the descriptions by Rouiller (Fig. 5).[26]

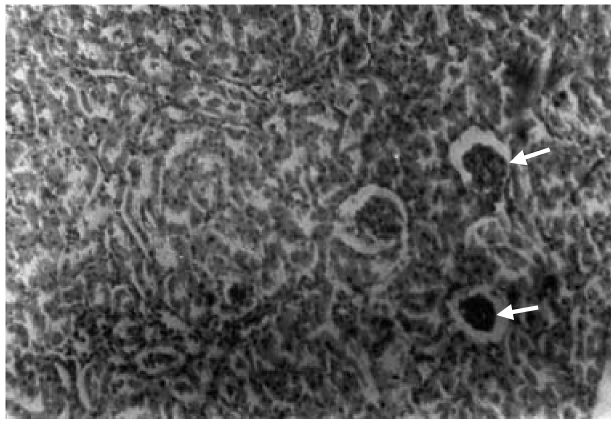

Fig. 2 Atrophic glomerulus can be seen (marked by arrows). The tubular epithelium looks swollen and it has obliterated the lumen.

DISCUSSION

From proximate analysis, the moisture content of *H. indicus* root powder, from plants collected during the monsoon season, was more than that of other powder samples of plants that were collected during other seasons. This can be caused by the increase in humidity. There is not much difference in these values of moisture content as compared to the earlier published values.[27] Similarly, fibre content, acid insoluble ash, total ash, and water soluble extractive values did not differ from the reported values.[28,29]

A cold extract showed a greater number of phytochemical bands on the HPTLC plate as compared with the soxhlet extract. This may be caused by the degradation of phytochemicals. The methenolic extract of root powder was found to be stable for a period of 2 days; this was observed from the chromatographic data. These findings suggest that the extract should not

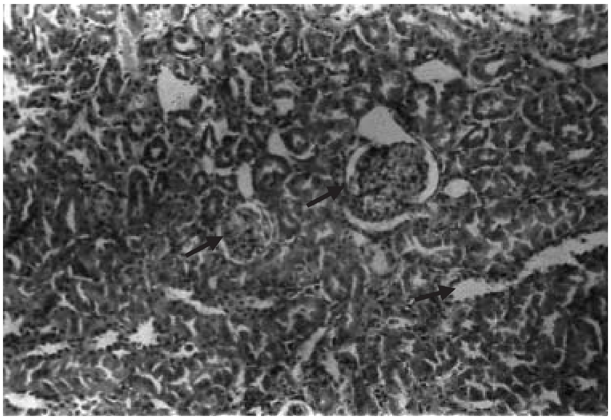

Fig. 3 Atrophic glomerulus can be seen (marked by arrows). The changes are of a moderate intensity.

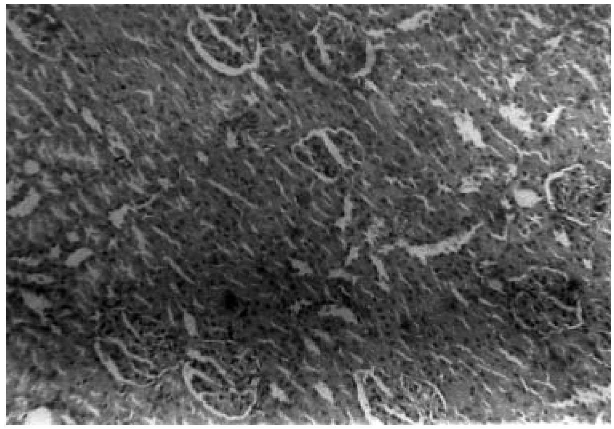

Fig. 5 Normal control group showing no atrophic glomerulus or lesions.

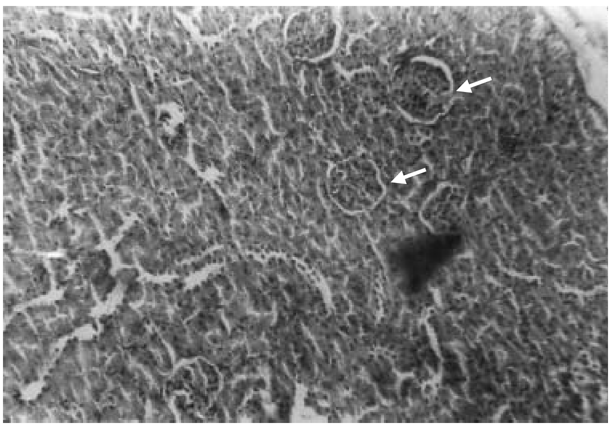

Fig. 4 Lesions of glomerulus, tubular epithelium were graded as being mild to moderate.

be used for more than the period specified. The degradation was not significant, with the concentration of the components still being high up until the fourth day. Significant degradation was observed from first week onwards, and, by the end of third week, the concentration of the three phytoconstituents was almost negligible.

Animal Studies

Toxicity study

From the toxicity study results, the drug was found to be relatively safe as no mortality was noted even when the highest dose was used. There were no changes observed with respect to the general behaviour, food and water consumption and bodyweight changes of the rats.

Efficacy study

Studies with pharmacological agents such as calcium channel blockers and angiotensin-converting enzyme (ACE) inhibitors have raised a new hope for patients. A 'nephroprotective' role as their effects has been suggested.[30] However, studies carried out on calcium channel blockers by Gomez *et al.*[31] and Biachini *et al.*[32] could not prove these claims. Similar disappointment was felt when ACE inhibitors were evaluated by Zucchelli *et al.*[33] and Mourad *et al.*[34] On one instance, ACE inhibitors have proved to be nephroprotective in diabetic nephropathy, but in contrast, various clinical studies in non-diabetic diseases are inconclusive.[35] Thus, their role as nephroprotective agents not only requires confirmation, but hurdles of cost, side-effects and contraindications also need to be overcome. Hence, we conducted the present study on the herbal drug *H.indicus*, which has been mentioned in the literature to be useful in the treatment of kidney disorders.[1,2] To the best of our knowledge, this is the first study providing information on the renoprotective action of the herbal drug, *H.indicus*, in gentamicin-induced renal dysfunction rats. The dose and route of administration of this test drug in rats was extrapolated from the dose used in humans in clinical practice, where it was given in combination with other drugs.

A dose of 90 mg/kg bodyweight was selected, which was injected i.m. for 6 days. In this group of rats (gentamicin control, gentamicin natural recovery and gentamicin plant recovery), there was a fall in bodyweight. The progressive rise in BUN with gentamicin administration indicated a decrease in the functional capacity of the kidneys because of acute renal failure. Raised levels of creatinine substantiated these findings. Gentamicin-injected rats from the present study demonstrated the typical pattern of aminoglycoside nephrotoxicity, characterized by polyuric renal failure, proteinuria and necrosis of proximal tubular epithelium as reported in a study by Kronin *et al.*[36] On autopsy, the kidneys appeared pale and swollen, with a rise in kidney weight being characteristic of damage. The findings were confirmed when histopathology was conducted. It showed a loss of basement membrane, tubular necrosis and casts in tubular lumens, all of which are typical of kidney damage.

Hemidesmus indicus prevented the fall in bodyweight of rats, which was seen in other groups that received gentamicin. The levels of BUN and creatinine showed a rise after 6 days, which was also seen in the plant-treated group, but this was less than that observed in the group receiving gentamicin (Table 7). These parameters indicated preservation of functional capacity. The attenuation of damage was confirmed on histopathology. In general, normal architecture was preserved (Fig. 4).

The uptake of gentamicin occurs after filtration into the renal tubule from the luminal side. It is transported across the lipid membrane by pinocytes and gets concentrated within the cell lysosomes.[37] It may impair the lysosomal catabolism of phospholipids by inhibiting their hydrolysis. Myeloid bodies may appear as the complex polar lipids accumulate. Cell damage could be mediated by the inhibition of mitochondrial oxidative respiration or by changes in the mitochondrail cation,[38] permeability or because of free radical generation.[39]

It is also known that gentamicin may impair the lysosomal catabolism of phospholipids by inhibiting their hydrolysis. Gentamicin has also found to appropriately halve ATP/ADP ratios (because of increased ADP), including a drug-induced defect in cellular energetics.[40] Thus, ATP depletion proceeds over cell necrosis in gentamicin nephrotoxicity, suggesting that energy-generating mitochondria are an important intracellular target for these drugs.[8] *Hemidesmus indicus* might be acting at any of the cellular levels. It might have prevented gentamicin uptake in cells or could have counteracted the effects of gentamicin inside cells. Further evaluation on the actual mechanism of action of the drug needs to be evaluated by using different methods such as therapeutic monitoring of drug levels in serum, which can be done by using microbiological assays or by rapid methods such as radioimmunoassay (RIA) and acetylating radio-enzymatic assay (ARA).[41] Tissue gentamicin levels can be measured, and from the decrease in the uptake of tissue gentamicin, the action of the drug can also be evaluated. As explained earlier for a number of nephrotoxic reactions, free radical-induced damage is a prime mechanism of injury. Hence, to estimate the generation of free radicals in animal tissue, other tests like MDA (malondialdehyde)[42] can be measured as a marker of lipid peroxidation. The prevention of a rise in MDA levels would indicate that a plant shows anti-oxidant properties, and anti-oxidant compounds have shown renoprotective effects.[43] Furthermore, pharmacokinetic studies in serum and tissues also need to be conducted where the time of drug absorption into tissues, as well as the serum levels, can be determined.

After the oral administration of *H. indicus* to rats, lipid levels have reduced, and this has been shown in many experimental models.[11] The plants might have hypolipidaemic activity, and if this is the case, this needs to be further evaluated in future studies. Previously, published work also states that the root derived from the cell culture of *H. indicus* was found to be effective as it prevented hypercholesterolaemia in experimental rats.[11]

In the present study, treatment with the plant root powder caused an increase in haemoglobin level in rats, which may be caused by the high percentage of iron content in the plant; this also requires further investigation. A reduction in the serum biochemical and enzyme levels, proteinuria and other parameters studied for plant root powder-treated rats as compared to the gentamicin control and recovery rats suggests the potential of the use of this plant in the treatment of gentamicin-induced nephrotoxicity.

The treatment of renal toxicity with standardized plants help in the management of renal impairment, which is induced by gentamicin in rats. It is evident that the development of quality controlled standardized herbal medicines, along with the adoption of good manufacturing practices and validation of claims of therapeutic efficacy, are the challenges in the years ahead if the resurgence of global interest, which is obvious today, in these drugs is to be sustained for a considerably long period of time in the future. The present work is an effort in this direction and it proves the efficacy of *H. indicus* in the management of nephrotoxicity induced by aminoglycosides such as gentamicin. This plant shows promise as an adjuent in therapy using aminoglycosides.

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